

The structure of transaction record (.msg file)

Each record will take 100 bytes in message file and appends to the file tail one by one.

.Msg File Offset	Field	Descriptions	Database Table Field of "eventlog"
0	Length	0x21	
1	Event code	Condition code of record, please refer to the following table:	LOG_CODE
2	SID	Source Node Address	CTL_NODE
3	Second		TM_EVENT
4	Minute		
5	Hour		
6	Week	1-7=Sunday ... Saturday	
7	Day		
8	Month		
9	Year	00-99 stand for 2000-2099 year. Fix 65h for AR721/727H	
10	Reader Node	The RS485 Node ID of Reader under Controller Panel (AR716E)	SUB_NODE
11	Card Addr-H	User index Bit 15~08. (Tag ID Bit [15~08] for invalid card event **)	USER_ADDR
12	Card Addr-L	User index Bit 07~00. (Tag ID Bit [07~00] for invalid card event **)	
13	Sub Code		LOG_SubCode
14	Sub Function		LOG_SubFunc
15	Ext Code		LOG_ExtCode
16	User Level		LOG_Level
17	Ext Data – H	Tag ID Bit [15~08]	TAG_UID
18	Ext Data – L	Tag ID Bit [07~00]	
19	Door Number	Door Number of the reader	DOORNO
20		Rev	
21		Tag ID Bit [31~24]	TAG_UID
22		Tag ID Bit [23~16]	
23		Transfer Value Bit [15~08]	TAG_PAY
24		Transfer Value Bit [07~00]	
25		Tag Balance Bit [15~08]	TAG_BALANCE
26		Tag Balance Bit [07~00]	
27			
28			
29			
30			
31			
32		Controller Node ID of 701Client force control	
34		Controller Sub-Node ID of 701Client force control	Note *1
35			
80~81		0xAA.0x55 Mark for 701Client log message	
82~87		IPv4 of AR727CM and TCP Port	
92		Area Code	AREA_CODE
93		IPCam Capture Flag, 0xA0~0xA9, 此欄位不為0表示這筆訊息可有配對影像檔.但是序號只取Low nibble.	LOG_FLAG
94		0x72. ** New version AR721 log that including tag UID	CTL_TYPE Note: *2
95		0x71. ** New version AR721 log that including tag UID	
96	Node ID	Node ID of the controller or multi-door controller (Field 10 is sub node controller).	---
97	Message type	64 mask for Standard SOYAL message, 0~63 is Node of MCU992 0x00 mask for system message. For login/logout message (Function code:20/21) the buffer[12~41] is operation user name and buffer[96] is user index in system.	---
98	LAN Base	0: system message. (MCU992) 1: the message from the AR-Serial controller. (AR-serial) 100 Not MCU message	LOG_TYPE
99	Confirm	0: Never(MSG_LIST_ALARM), 1:Confirmed(MSG_LIST_ALARM_CONFIRMED)	

NOTE 1: The Blue Color Filed will keep in "LOG_BUFFER" of Table "eventlog" in Database Mode.
NOTE 2: The CTL_TYPE Code is defined in "Soyal Controllers ID.pdf".

Note:

**: For old version controller only. (716E 8V9, 829E 7V6, 721H3V5 and before)

1. If message type(field 98) equal to 0x00, that means this message is user login/out(0x20/0x21 at field 10) and field 96 will show system user index.
2. Sub Code have total 8 Bits. Among Bit7~5 stand for duty code.
 000xxxxxB: on duty, 001xxxxxB: off duty, 010xxxxxB: overtime in,
 011xxxxxB: overtime out 100xxxxxB: break out, 101xxxxxB: break in,
 110xxxxxB: go out, 111xxxxxB: return

Controller Function code define		
0	Site Code Error	
1	Access via User Address and PIN or PIN Only Error	
2	Key Pad Locked	
3	Invalid Card Access	
4	Access time zone is error	
5	Access door group is error	
6	Expire date	
8	User input PIN code incorrect	
9	Duress Message. (Emergency Help)	
10, 11	Normal Access via car	* 1
12, 13	Force Controller Relay On/Off [12/13]	
14	Enter Arming status	
15	Enter Disarming status (Exit Arming)	
16	Door open via Egress Button	
17	Alarm event, Set bit7 of Sub Function to indicate force open alarm.	
20	Controller power off	
21	Duress event	
22	Help message of guest	
23	Cleaner access message	
24	Controller power on	
25	Force Controller Relay On/Off failed	
26	Return to normal status (Analog I/O event)	
27	Panic Button Event	
28	Normal Access via PIN code	* 1
39	Normal Access via fingerprint	* 1
29	DI status has changed,SubCode:00:Off, 01:On. Reader Node ID: DI number	
30	Anti-pass-back error	
31	Reader disconnected with controller	
32	Reader reconnected with controller	

33	User change PIN code	
34	User change PIN code error	
35	Controller Enter Auto Open Procedure	
36	Controller Exit from Auto Open Procedure	
37	Disarming by auto time procedure	
38	Arming by auto time procedure	
39	Access by Fingerprint	
40	Fingerprint image failed	
41		
42	Remote Key: Up	
43	Disable Reader	
44	Enable Reader	
45	Remote Key: Panic	
46	Entrance	
47	Appearance	
48	Trigger counter	
49	Reader Relay Latch On/Off. Sub Function:00:Door Relay ON,01:Door relay OFF	
50/51	DI point Close/Open(50/51), The Pp number put in 【sub-function】	
52		
53	Enter/Exit Edit Mode	
54	Remote controller message	
55	Global free access (Controller in global free mode)	
60	Flash Card But No Open Door	
63	Soyal Open System Rule : Instruction Failed	
64	Soyal Open System Rule : Deduct value failed	
65	Soyal Open System Rule : Global Tag access ok	
66	Soyal Open System Rule : License layer feiled	
67	Soyal Open System Rule : Before begin date	
68	Soyal Open System Rule : Expiry date	
69	Soyal Open System Rule : Balance not enough	
70	Soyal Open System Rule : Deduct value ok	
71	Soyal Open System Rule : Read floor data from tag failed	
72	Soyal Open System Rule : Global access tag deduct value success	
73	Soyal Open System Rule : Global access tag balance not enough	
74	Soyal Open System Rule : Global access tag deduct value failed	
75	SOR global access ok without deducted	
86	Black table tag accessed	

[illegible]

200	Interphone message	
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System Function code define		
17		
18		
19		
20	User Login	
21	User Logout	
22	Controller Offline	
23	Controller Online	
24	Force Open Door by Remote User	
25	Force Open Release by Remote User	
26	Fire Alarm triggered from AR727CM-IO	

Communication protocol:

Transmission rate		9600,N,8,1 (Only for UART)				
Data format		Binary HEX "hexadecimal"				
Data package						
Head 8bits	Length 8bits	Destination ID 8bits	Function Code 8bits	Data ...	XOR 8bits	SUM 8bits

	Size		XOR (0xFF)	SUM (0x00)
Head	1	0x7E	No	No
Length	1	Data length Indicator. Denotes the length from Destination to SUM.	No	No
Destination ID	1	Destination Node Address. 00: Reserved for the bus master (Controller) FF: Broadcast to each reader.	Yes	Yes
Command	1	Instruction command.	Yes	Yes
Data	00 ~ D0	The length of Data Block depends on the instructions.	Yes	Yes
XOR	1	XOR each byte from [Destination ID] until last data byte with 0xFF.		Yes
SUM	1	Summing each byte from [Destination ID] until XOR byte with 0x00		

Ex: Polling status from Node 1.						
Head	Length	Destination	Command	Data ...	XOR	SUM
0x7E	0x04	0x01	0x18		0xE6	0xFF

The Data Package format from Remote Controller

Package			7E 16 [0A] 01 [19] [35] [0D] [02] [14] [0C] [04] 01 [AA 9F] 00 00 00 00 [11 77] 01 04 9C FF
Offset		Name	
0	7E	Header	Header Code of each data transmits
1	16	Length	There have 0x16(22) bytes followed.
2	0A	Event	0x0A: Flash Card(Tag)
3	01	Source	Node address of source reader
4	19	Second	
5	35	Minute	
6	0D	Hour	
7	02	Weekday	Start from Sunday to Saturday. (1,2,3...7)
8	14	Day	Day of month
9	0C	Month	
10	04	Year	
11	01	Source	Node address of source reader
12	AA	UID1	User address or Tag ID (Bits 15~08) if invalid tag access
13	9F	UID0	User address or Tag ID (Bits 07~00) if invalid tag access
14	00		Reserved
15	00		Reserved
16	00		Reserved
17	00		Reserved
18	11	UID3	Tag ID (Bits 31~24)
19	77	UID2	Tag ID (Bits 23~16)
20	01		Door Number of Reader, For 828E/727H/721H(0xF0/0xF1->829/Ext Reader)
21	04		Reserved
			Tag ID bit 15~08
			Tag ID bit 07~00
			Transfer Value bit 15~08 (SOR only)
			Transfer value bit 07~00 (SOR only)
			Balance Value bit 15~08 (SOR only)
			Balance value bit 07~00 (SOR only)
		XOR	
		SUM	

- 1) The remote site will connect to server site when the message buffer of remote controller is not empty.
- 2) After connect to server site the controller will send message to server and delete the message if transmit successes in TCP/IP link.
- 3) When all message have send out then the remote site will disconnect automatically.
- 4) Any other require ...